S Lab Sync (Python) — System Architecture

How the app is put together, from launch to the MySQL database

User runs python main.py
(or double-clicks
run_slab_sync.bat on Windows)

main.py — App.startup()
Loads the saved MySQL details,
or shows the setup dialog

db_setup_dialog()
(first run only) — asks for
MySQL host / user / password

ConfigStore (db.py)
Remembers the connection in
~/.labmanager/python_db.json

App.login()
Master login — default Master / 7777
(changeable in the app)

main.py — Tkinter Interface Layer (class App)

theme.py

light / dark
palettes &
ttk styles for
every screen

App.build_ui()
Sidebar navigation + Light / Dark toggle

Overview
page

Students
page

Teachers
page

Equipment
page

Chemicals
page

Experiments
page

db.py — LabDatabase (data access, one class for all tables)

count / fetch_all /
fetch_one

insert / update /
delete

experiments_overview()
(dashboard query)

set_experiment_students()
student_ids_for_experiment()

_require_table()
table-name whitelist

db.py — Connection, Schema & Local Settings

LabDatabase.connect()
One short-lived connection
per operation

initialize()
Creates lab_management
on first run, upgrades old DBs

CredentialStore / ConfigStore
Master login and MySQL
details as local JSON

tests/ + CI

67 automated tests

theme contrast
credentials storage
DB logic
UI renders in both themes
real MySQL integration

GitHub Actions runs them
on every push

checks

mysql-connector-python (MySQL driver)
installed with: pip install -r requirements.txt

~/.labmanager/
credentials.json • python_db.json

MySQL Server
database: lab_management
teachers • students • equipment • chemicals • experiments • experiment_students